WHY GROUNDING

Grounding is the difference between an answer and a guess.

Grounding is the agent using retrieved evidence in the answer, with citations that survive a second look. Without grounding the model improvises from training — usually plausible, sometimes wrong, never traceable.

CASE · WITHOUT GROUNDING

A confident improv

USER ASKS

“When does my refund post?”

Refunds typically post within 5–7 business days. Always mark your refund as pending before submitting a duplicate. Processing time depends on your region and bank.

sounds right · not verified · not from your data

CASE · WITH GROUNDING

A traceable account

USER ASKS

“When does my refund post?”

Refunds post within **7 business days** for card refunds and **10 business days** otherwise — per Refund-Policy-v4.pdf.

↑ Refund-Policy-v4.pdf, p.2

GROUNDING IS NOT “SOUNDING SMART”

Grounding is the answer being traceable. Same fluency, different floor.

SOURCES · THE LANDSCAPE

Four families of knowledge sources feed an agent — each with its own rules.

A **knowledge source** is any material the agent is authorised to read at answer-time: sites it owns, sites it doesn't, structured tables, and indexed documents. The source is decided at design time; the answer at run time.

01 · SITES YOU OWN

SharePoint

Sites, libraries and pages. Permissions ride along automatically — the agent only returns what the user could read.

02 · STRUCTURED TABLES

Dataverse

Tables you author or already have. Up to 15 tables per source, 2 sources per agent.

AGENT

The agent

03 · PUBLIC SITES

Public websites

Anchored URL lists the agent is bound to. Auth-walled pages are out of reach.

04 · INDEXED CORPUS

Azure AI Search

A search index you control. Vector + keyword + filters, semantic ranker optional.

Choose the source that matches the **trust boundary** the answer must respect — not the source easiest to wire.

SOURCE FAMILY 01

SharePoint — the site becomes the source.

A **SharePoint knowledge source** is one or more sites, libraries, or pages the agent is bound to. The indexer picks up files at design time; access checks re-run at answer time.

WHAT GETS INDEXED

Site → library → file → passage

The path narrows with each step. Citations sit at the passage level.

PERMISSIONS

Honoured at answer time

A user sees only what they could already open. Inheritance works; explicit deny wins.

FILE TYPES

.docx, .pdf, .pptx, .md, .txt

Lists are supported as structured sources; items inherit list permissions.

WATCH FOR

“Files” isn’t a knowledge source until it’s permitted

Files uploaded during agent creation live in Dataverse file storage, not SharePoint. They’re convenience storage; for canonical answers, point the source at the real site.

SETUP AT A GLANCE

Five minutes, three knobs.

STEP 01

Add a SharePoint site or library URL.

STEP 02

Confirm scope: all files, or a folder / list filter.

STEP 03

Pick sync: one-shot, or scheduled refresh.

CAVEAT

A user who can’t open the source can’t get an answer from it. The agent never widens access; it only narrows.

SOURCE FAMILY 02

Dataverse — tables the agent already knows the shape of.

A **Dataverse knowledge source** is a curated set of tables you author or already own. The agent queries them like a database and answers with rows behind it.

WHAT GETS BOUND

Up to 15 tables per source, 2 sources per agent.

case · product catalogue

sku	name	price
A-1041	Alloy Mug	$18
B-2077	Glass Carafe	$44
C-3302	Walnut Board	$62

WELL-SUITED

Lookups, statuses, prices, owners — data that already has columns and rows.

VS. SHAREPOINT SOURCES

Dataverse binds to rows and columns you queried, not files in a library. Use it when the question has a precise shape, not a paragraph to summarise.

TWO CAUTIONS

Author the right tables, then leave the agent room to filter.

Pre-curated sources answer sharp questions; over-broad tables drag in noise. If you need prose, point at a knowledge article instead.

WATCH FOR

Row-level security rides along

If the user can’t read the row, the agent can’t reveal the row. Test with a low-permission user, not your admin.

SOURCE FAMILIES 03 & 04 · THE LOOSE COUSINS

Public sites and uploaded documents — useful, but with caveats.

Both sources **look** like SharePoint and feel like SharePoint, but neither inherits SharePoint permissions. They are scoped to the agent, not to the directory.

PUBLIC WEBSITES

WHAT IT DOES

Crawls up to N anchored URLs the maker lists.

BEST FOR

Docs you ship but don't own — vendor reference, public KBs, regulatory text.

Pair with content moderation; the public internet is not curated for you.

WATCH FOR

Auth-walled pages fail silently. The agent doesn't error — it just answers from what it could reach.

UPLOADED FILES

WHAT IT DOES

Holds files you uploaded during agent creation.

`PDF · DOCX · PPTX · TXT · MD · XLSX`

BEST FOR

Small, fixed reference sets the agent owns outright.

A short FAQ, a single spec, a glossary — things small enough to live with the agent.

WATCH FOR

Files live in Dataverse file storage, not SharePoint. Convenience, not governance.

PICKING A SOURCE · A FLOW, NOT A VIBE

Pick the source by what the answer must respect — not what's easiest to wire.

A decision **short-circuits to one rule**: where does the truth live, who must be allowed to see it, and how big is the body. Three questions, four answers.

QUESTION 1

Where does the truth live — inside or outside the tenant?

QUESTION 2

Is the answer rows in a table or prose to summarise?

QUESTION 3

Do you need filters, faceting, or a custom ranker?

WHEN IN DOUBT

SharePoint, almost always

Honours permissions, lives with the rest of the work, citations behave.

WHEN ROWS WIN

Bind to Dataverse

Status, lookup, owner, price, count.

WHEN OUTSIDE THE TENANT

Public websites

Vendor docs, public KBs. Whitelist URLs; revisit at refresh.

WHEN IN SCALE & CONTROL

Azure AI Search

Custom ranker, filters, faceting.

RULE OF THUMB

Match the source to the trust boundary, not the convenience.

If permissions don't ride along, neither should the answer.

RARE BUT LEGITIMATE

A few uploaded files

Glossary, single spec, agent-specific fixed scope. Not the canonical home.

CITATIONS · WHERE THE ANSWER CAME FROM

A citation is a finger pointing back at a passage.

A **citation** is the small reference attached to a fetched passage: source, link, sometimes a snippet. Citations make the answer **legible**; without them the user is asked to trust tone.

A MOCK GROUNDED ANSWER

Refunds post within **7 business days** for card refunds [1] and **10 business days** for refunds over $500 [2]. An exception can be requested via the portal [3].

↑ [1] Refund-Policy-v4.pdf · p.2 ↑ [2] Refund-Policy-v4.pdf · p.4 ↑ [3] /help/refunds/exceptions

FOUR PARTS

Only some are required.

TITLE

The doc or page the passage came from.

LINK · REQUIRED FOR TRIAGE

A URL the user can open in one click.

SNIPPET

The exact passage the model lifted.

LOCATOR

Section, page, paragraph. The closer, the better.

WITH AZURE AI SEARCH

The index schema needs a URL field. Without it your citation row comes back empty — and looks uncited.

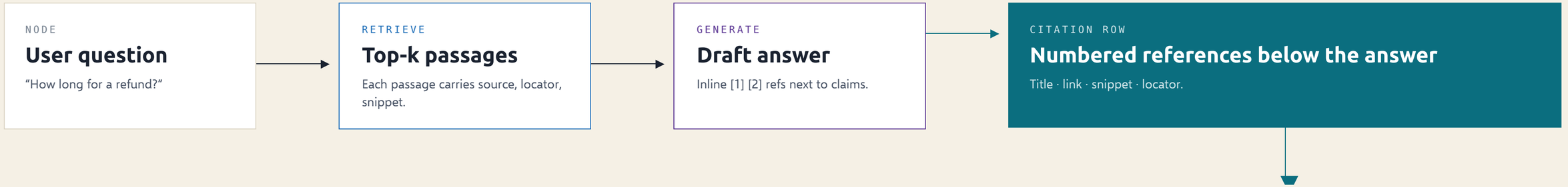
A CITATION IS METADATA, NOT PROSE

It earns the right to be in the answer. A weak citation is worse than none — it borrows trust it hasn't repaid.

CITATIONS · THE WIRING

Every citation is a thread the user can pull to verify.

A grounded answer is a flow: the model writes a sentence, the citation is the matching span in the source. If the wire is broken the sentence reads — but trust doesn't survive.



WHEN THE WIRE IS BROKEN

Three quiet failure modes you'll see in production:

· **Citation without snippet** — looks like magic; user re-checks trust.
· **Snippet that doesn't back the claim** — worse than no citation.
· **Link returns 404** — the agent will be cited as untrustworthy long after it was fixed.

THE SOURCE

The passage, page or row sits behind the marker.

When the user clicks [2], the same passage they could have read themselves. The wire from snippet to source is the contract.

A citation is a small piece of **repaid trust**. Test every answer against a stranger who has never heard of your agent.

RAG · THE SERIOUS INDEXER

Azure AI Search is the indexer Copilot Studio leans on.

Azure AI Search is a managed search index that lives next to Copilot Studio. It is the right tool when the corpus is large, when you want vector + keyword, or when filters and faceting earn the answer.

WHY IT SITS ALONGSIDE

Copilot Studio handles conversation. Azure AI Search handles retrieval.

Bind an agent to an index schema you control; the agent queries it at run time and wires citations through it.

WHAT YOU KEEP

- Index schema
- Vector + keyword config
- Filters & semantic ranker
- Schedules & indexing profile

WHEN IT EARNS THE SLOT

Corpus is large or messy

Tens of thousands of files, multiple formats, duplicate topics, retirement churn. Native sources can’t keep up cleanly.

OR

Hybrid search wins

Vector + keyword + filters in one query.

OR

Citations are critical

A controlled URL field per passage, ready to render.

THE CONTRACT

Index lives in Azure. Agent talks to it over a connection.

The index is yours to re-rank, refresh, or replace. The agent doesn’t own the corpus — it borrows rows of it on demand.

RAG · INGESTION PIPELINE

A pipeline turns the corpus into rows the agent can query.

Six stages run in order. Each produces a row, log, and a checksum. Skimping the middle three is where most failures li

01

Discover

Pick sources: blob, SharePoint, SQL, API.

02

Extract

Pull text out of PDFs, DOCX, HTML, scans (OCR).

03

Chunk

Slice into passages of 200–800 tokens with overlap.

04

Embed

Turn each chunk into a vector; pick a model once.

05

Index

Write the rows: vector, fields, URL, locator.

STEP 06

Refresh on a schedule

The pipeline runs again on the timer you set; stale rows retire by id.

WATCH FOR · CHUNKING

A chunk mid-table loses its meaning. Chunk by section heading when you can; never slice a heading off from its content.

WATCH FOR · SCHEMA

Add a URL field and a title field. The citation row lives or dies on those two columns existing.

RULE OF THUMB

If you can’t refresh, you can’t ship. A static index is a future incident report.

The ingestion pipeline is a **diary** of decisions. When it disagrees with you, the answer feels stale — the index usually is.

RAG · QUERY-TIME FLOW

Retrieval is five steps; three can change the answer.

Every answer goes through this flow. Tune **rewrite**, **search**, and **rerank**; the other two are usually fine out of the box.

STEP 01 · TUNE THIS

Rewrite

Expand the question for retrieval: clarify “it”, drop fillers, fix typos.

STEP 02

Embed

Turn the rewrite into a vector with the same model used at index time.

STEP 03 · TUNE THIS

Search

Hybrid query: vector + keyword + filters from the same index schema.

STEP 04 · TUNE THIS

Rerank

Trim to top-k–1; re-order by relevance; cut near-duplicates.

STEP 05

Generate

Write the answer with citations wired to spans.

RESPECTS NOTHING

Embed — pick once, never change mid-stream. Mismatched models = nonsense results.

RESPECTS PERMISSIONS

Search — filters prune out content the user can’t read; tag ACLs before they ever flow back.

RESPECTS THE USER

Rerank — what looks most-real can still be wrong; trim duplicates, keep three to five real candidates.

WHERE ANSWERS USUALLY FAIL

Step 01 is half the win. Step 04 is the other half.

A bad rewrite retrieves the wrong rows. Rerank without trimming keeps weak candidates alive. Skip those, and the model has nothing to write a great answer from.

HYBRID · THREE MODES, ONE INDEX

Sometimes the question is about meaning; sometimes about a word.

A **hybrid search** query runs more than one retrieval mode against the same index and merges the rows. The right mix depends on what the user actually asks.

VECTOR SEARCH

Same idea, new words

Matches by embedding distance — paraphrases, synonyms, vague questions find their way home.

WINS WHEN

“how do I undo a shipment that already left?”

LOSES WHEN

“SKU-A-1041” — a code with no semantic neighbours.

KEYWORD SEARCH

Same word, exact spelling

Token-based match. BM25-style scoring; ranks lexical hits first.

WINS WHEN

Error codes, IDs, regulator names. Exactness wins; flags in tokens matter.

LOSES WHEN

The user described the policy in their own words.

HYBRID SEARCH

Both, with filters

A fused score over vector + keyword; optional semantic ranker on top. Most production agents live here.

WINS WHEN

Wide question-tariff, with filters on region, product, or date.

COST OF BEING RIGHT

Tune weights; the wrong mix recalls neither rhetorical nor exact questions cleanly.

A PRACTICAL DEFAULT

Start with hybrid and a semantic ranker. Specialise to vector or keyword only when an evaluation set proves the gap.

SCOPING · THE TRUST BOUNDARY

An agent speaks for what a user is already permitted to read.

Scoping is the set of rows the agent is allowed to look at, on behalf of the user asking. **Access-control** is the check that runs before a row reaches the answer. The same idea, repeated in two places.

LAYER 01

User

Authenticated, identified

Row-level filter

Passage / row arrives

LAYER 02

Directory permissions

SharePoint sites, lists, file ACLs enforce the first wall.

LAYER 03

Row-level filter

Dataverse row security, indexed “owner” field, geography tags.

LAYER 04

Only what passed arrives

Filtered rows are what the model ever sees; nothing else is searched.

SCENARIO A · OVER-PERMISSIVE

Source lists the whole library — the agent answers from a doc the user can’t open. The reference itself reveals a permission gap.

SCENARIO B · UNDER-TAGGED

Index row has no ACL field. Filters fall back to default-block; everyone sees nothing.

SCENARIO C · THE RIGHT SHAPE

Index carries owner + geography; tests run with a low-permission user before publishing.

Test the agent under the hardest accounts you have, not the friendliest.

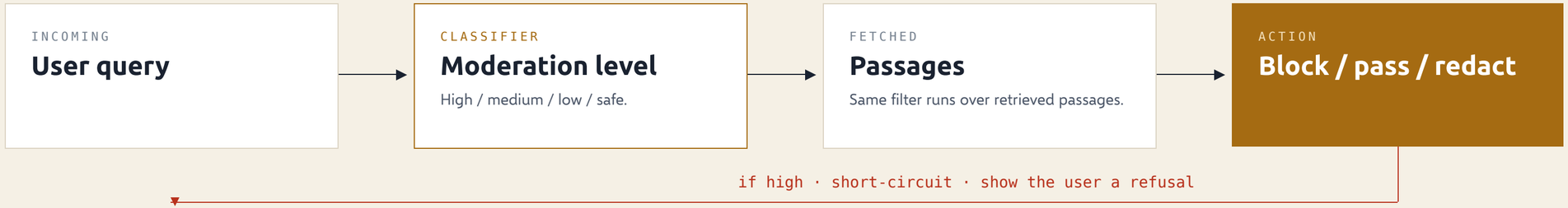
Permissions are a feature of the source, not a switch on the agent.

Choose the source, and you’ve already picked the trust boundary.

WHERE ACCURACY BREAKS

Moderation filters input; failure modes live in the output.

Content moderation classifies queries and passages against a policy (safe, low, medium, high) and short-circuits the unsafe ones. It guards entry; it does not guarantee accuracy.



FOUR SILENT FAILURES · THE PART MODERATION DOESN'T CATCH

FAILURE 01 · UNSUPPORTED CLAIM

Sentence has no citation behind it. Printed tone, no proof.

FAILURE 02 · MISMATCHED SNIPPET

Citation present, but the snippet doesn't actually back the claim.

FAILURE 03 · OUTDATED REFERENCE

Citation points at last quarter's draft. Pipelines without refresh always betray this.

FAILURE 04 · PERMISSION LEAK

A row retrieved across an ACL boundary — rare, but lethal.

Moderation guards the door. Accuracy lives downstream — tune both, in order.

LAB · BUILD A RAG PIPELINE

Build a multi-source agent, then grade it on a stranger.

Eight steps over ~70 minutes. The last three aren’t optional — they’re how the lab earns its keep.

01 — 8 MINUTES

Pick a topic and a reader

One persona with one job. “HR answers a contractor about leave”.

02 — 8 MINUTES

Add a SharePoint source

Bind to one site with a known doc set. Confirm indexing completes.

03 — 10 MINUTES

Add Azure AI Search

Bind an index with a URL field and title field. Hybrid + filters on.

04 — 8 MINUTES

Ground with citations

Toggle citations on. Verify a citation appears next to a real claim.

05 — 10 MINUTES

Tune rewrite and rerank

Top-k–1 to start. Throw five weird questions; study the rerank.

06 — 12 MINUTES

Test on ten real questions

Five for sources you have, five you don’t. Mark pass / fail / wrong-citation.

07 — 8 MINUTES

Set a refresh cadence

Schedule re-indexing. Log when each row last touched.

08 — 10 MINUTES

Run the lab with a low-perm user

Invite a stranger, watch what they see, write down the gaps.

SUCCESS SIGNALS

What “done” actually looks like.

SIGNAL 1

10/10 questions get a citation

Even “I don’t have that” counts, when paired with a partial-citation track.

SIGNAL 2

Citations open the right row

Not a sibling, not a parent — the actual passage.

SIGNAL 3

Low-perm user can’t see a gap

Try a user without rights to the source; the answer degrades gracefully.

SIGNAL 4

Index refresh runs on a timer you trust

Not a one-shot. Logs say when each row last refreshed, in plain text.

THE RULE

If a stranger reads your citations and walks away unsure, the lab isn’t done.